Ecto's "Types and Casting" documentation[10] lists out all of the different field types supported. Field types are either primitives or custom types. These are some examples of primitive fields: :integer, :string, :utc_datetime_usec, :map, and more. You can even specify array fields using the {:array, inner_type} field type.

Ecto.Enum and Ecto.UUID are examples of custom field types. There are many different custom field types—and you can make your own—but you typically only need to use them for special cases.

Ecto types map to native types in the database that you use. A single Ecto type can map to multiple different native types. For example: varchar, string, and text PostgreSQL types are all handled by Ecto's :string type.

The field/3 function also accepts various options.[11] The most common option is :default, which lets you specify the default value for the column when null values are written or read from the database.

Another common option is :virtual. Use virtual fields when you want to attach custom data to a struct. You can add data to a virtual field either in Elixir—using Map.put/3—or from database queries.

Fields and associations go hand-in-hand. Let's take a look at associations next.

Schema Associations

Ecto schemas support one-to-one, one-to-many, and many-to-many associations. These association types have largely been standardized across database frameworks, so they will feel familiar. Here's how each type works:

One-to-One

The belongs_to/3 function defines a relationship that's either one-to-one or one-to-many. You use belongs_to/3 in the schema that contains the ID of the relationship. In our example, the SmsMessage schema includes a contact_id field.

The other side of the relationship uses the has_one/3 function to define the complete relationship.

One-to-Many

This type of relationship is similar to one-to-one, but the other side of the relationship uses has_many/3 to define the complete relationship.

10. https://hexdocs.pm/ecto/Ecto.Schema.html#module-types-and-casting
11. https://hexdocs.pm/ecto/Ecto.Schema.html#field/3